

Technical Document #753: Cloud Computing - Comprehensive Overview

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Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files. Terraform and CloudFormation are popular IaC tools.

Serverless computing allows developers to run code without managing servers. AWS Lambda and Azure Functions automatically scale based on demand.

Microservices architecture structures applications as a collection of loosely coupled services. Each service runs in its own process and communicates via APIs.

Containerization packages applications with their dependencies. Docker creates lightweight, portable containers that run consistently across environments.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Key Takeaways:

- Microservices architecture structures applications as a collection of loosely coupled services.
- Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet.
- Auto-scaling automatically adjusts compute resources based on demand.